

- LO1:** Identify the characteristics and components of academic research
- LO2:** Analyse research problems in various IT fields
- LO3:** Evaluate research activities for ethical/unethical conduct
- LO4:** Identifying different sources of information
- LO5:** Evaluate the quality/reliability of different sources
- LO6:** Apply referencing styles in the acknowledgement of sources
- LO7:** Evaluate different research strategies
- LO8:** Analysing a conference call
- LO9:** Evaluate different data collection and analysis methods for specific research scenarios
- LO10:** Understanding how the abstract is constructed

QUESTION 1: MULTIPLE-CHOICE [60]

1. Answer: A	[2]
2. Answer: D	[2]
3. Answer: D	[2]
4. Answer: D	[2]
5. Answer: A	[2]
6. Answer: B	[2]
7. Answer: D	[2]
8. Answer: D	[2]
9. Answer: C	[2]
10. Answer: A	[2]
11. Answer: C	[2]
12. Answer: A	[2]
13. Answer: A	[2]
14. Answer: D	[2]
15. Answer: A	[2]
16. Answer: B	[2]
17. Answer: B	[2]
18. Answer: D	[2]
19. Answer: B	[2]
20. Answer: C	[2]
21. Answer: B	[2]
22. Answer: B	[2]
23. Answer: B	[2]
24. Answer: A	[2]
25. Answer: B	[2]
26. Answer: C	[2]
27. Answer: B	[2]
28. Answer: B	[2]
29. Answer: B	[2]
30. Answer: B	[2]

QUESTION 2: TRUE/FALSE [10]

- 31. Answer: True: A [2]
- 32. Answer: True: A [2]
- 33. Answer: False: B [2]
- 34. Answer: False: B [2]
- 35. Answer: True: A [2]

QUESTION 3: DESCRIPTIVE QUESTIONS [30]

Answers may differ from one student to another. However, they must all lead to the same meaning.

1. What is the research methodology for this work? [3]

The research methodology involves:

Image recognition (to identify ingredients from user-provided images)

Convolutional Neural Networks (CNNs) (for processing and analyzing images)

Text-to-speech guided technology (to assist users in following recipes)

User-based testing (implied by the focus on user convenience and efficiency)

2. List five databases that can be used to identify the relevant articles for this study. [5]

Possible databases include:

IEEE Xplore (for AI and image recognition research)

ScienceDirect (for food technology and machine learning studies)

Web of Science (for nutrition and dietary-related research)

ACM Digital Library (for computer science and AI applications)

Scopus (for multidisciplinary research on smart food technologies)

3. What is the contribution of this study? [3]

The study contributes:

A recipe recommendation system based on image recognition (rather than text inputs). Integration of text-to-speech technology for easier recipe following. Additional features like grocery cart redirection for user convenience

4. What is the research problem for this study? [3]

The research problem is that existing recipe recommendation systems rely mostly on textual inputs or predefined ingredient categories, which may not accurately reflect what users have available in their pantries, leading to inefficiency and inconvenience.

5. Based on the abstract, what would be the potential topic for this study? [2]

"An AI-Powered Image-Based Recipe Recommendation System with Voice Guidance"

6. Why does this abstract not have in-text references? [2]

The abstract does not have in-text references because it summarizes the proposed study rather than a review of existing literature. It focuses on presenting the new approach rather than citing prior work.

7. List two keywords based on the topic you proposed in number five. [2]

"Image-based recipe recommendation"

"AI cooking assistant"